

Amiri Prescod

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WEBSITE: <https://amiriprescod.vercel.app/> | GITHUB: <https://github.com/aprescod12>

EDUCATION

Villanova University – College of Engineering, Villanova, PA

Master of Science in Biomedical Engineering | GPA: 4.00

Expected Dec 2027

Bachelor of Science in Electrical Engineering | Minor in Computer Science | GPA: 3.21

May 2026

ENGINEERING & PROFESSIONAL EXPERIENCE

Independent Biomedical Research, Villanova University

Fall 2025-Summer 2026

Research Assistant

- Acquired and processed biomedical signals to analyze human physiological and neurological behavior.
- Troubleshoot and restored functionality of a legacy sensor, improving data reliability for ongoing research.
- Developed a user-facing application to visualize and communicate real-time biomedical data.

The National Gas Company of Trinidad and Tobago Limited, Point Lisas, TTO

Summer 2025

Undergraduate Intern – Asset Maintenance

- Supported Electrical, Telecommunications, and Instrumentation teams in asset maintenance operations.
- Conducted site inspections, system troubleshooting, and operational diagnostics across industrial facilities.

Heft IQ, Remote

Summer 2024

Artificial Intelligence Intern

- Applied foundational knowledge of AI and machine learning to evaluate real-world use cases.
- Analyzed potential AI applications and translated concepts into practical implementations through guided projects

SKILLS

Engineering: Testing & Validation, Root-Cause Troubleshooting, Prototyping, Data Analysis, Technical Documentation

Programming & Embedded: Python, C, C++, SQL, MATLAB, ESP32, Particle Photon, Git, EasyEDA, Arduino, Git

Laboratory: Oscilloscope, Digital Multimeter, Function Generator, Soldering, Circuit Testing, Electronics Troubleshooting

PROJECTS

Medical Monitoring & Safety Device | <https://github.com/aprescod12/medical-monitoring-safety-device>

- Developed an embedded health monitoring system integrating fall detection, sleep-state analysis, and hardware-triggered emergency alerts.
- Designed real-time sensor processing and deterministic event handling for rapid safety response.

AFA Event Shift Sign-Up Platform | Django, PostgreSQL

- Built a role-based Django/PostgreSQL platform for shift sign-ups, waitlists, attendance, and reporting.
- Validated scheduling requirements, permissions, concurrency, notifications, and reporting through structured end-to-end testing.

Bat Tracking System Capstone Project | <https://amiriprescod.vercel.app/projects/baseball-bat-tracking>

- Integrated Azure Kinect and IMU sensing to capture bat motion and generate quantitative swing-performance feedback in a Virtual Reality setting.

Particle Photon Security System | <https://github.com/aprescod12/particle-photon-security-system>

- Developed a cloud-connected burglar alarm with motion detection and remote arming via Particle Cloud.
- Integrated signal filtering and time-based validation to reduce false positives in real-world conditions.

LEADERSHIP & ACTIVITIES

Incoming McLendon Fellow, Villanova Athletics

August 2026-May 2027

Academic Support & UNITAS: Community, Culture, and Engagement

NCAA D1 Student Athlete

October 2022-May 2026

Villanova University Track and Field | 20+ hrs/week

- NCAA D1 Sprinter; Team Leader, SAAC Representative, and International Student-Athlete Leader.

NSBE

August 2023-May 2026

NovaNSBE Executive Board Member (Pre-College Initiative Chairperson and Senator)

- Organized outreach events engaging local high school students in STEM education.
- Represented NSBE Chapter at conferences and voted for the National and Regional NSBE Executive Positions.